RCDEPS × Tesla Integration — Engineering Manual v1.0

Recursive Coherence■Driven Electrodynamic Propulsion System (RCDEPS)
Tesla■inspired Resonant EM Hardware, Measurement, and SOP

Executive Summary

This manual delivers a build■ready integration of Tesla’s resonant electromagnetic hardware concepts (coils, rotating fields, grounded towers) with the RCDEPS framework described in your uploaded documents. It specifies a coherent high■voltage emitter, a phase■locked multi■coil array, a Wardenclyffe■style grounded resonator, and a SpaceDrive■grade vacuum metrology stack. The design targets nano■ to micro■Newton forces with long■duration, artifact■controlled runs and includes a photon■thrust calibrator for truth■standard checks.

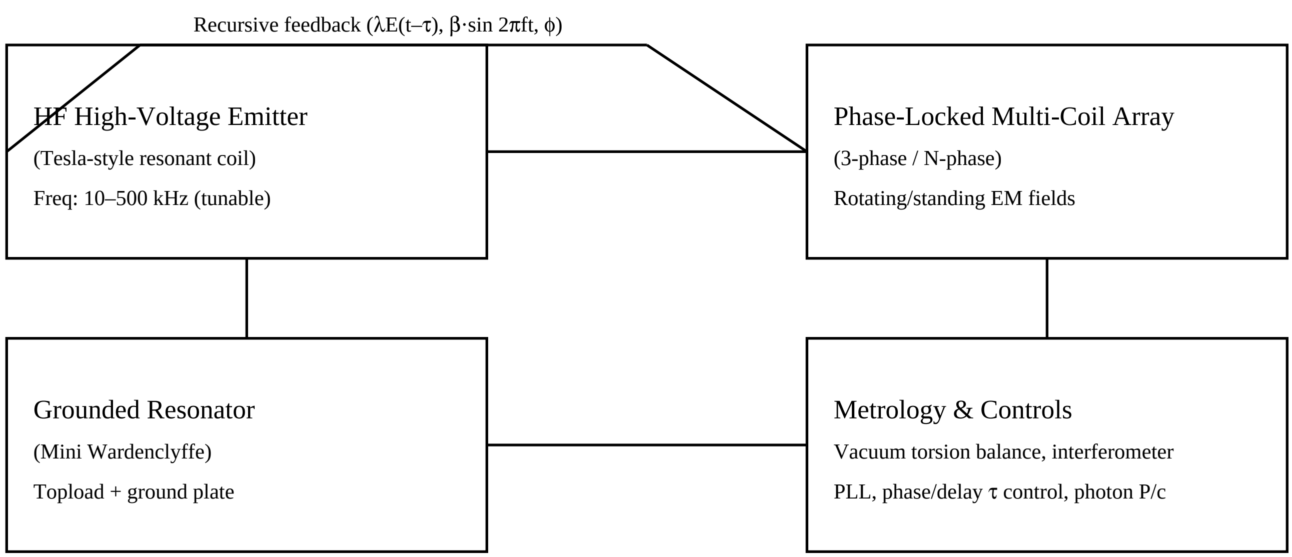
Performance Targets

• Baseline measurable thrust: 1×10■■ N (torsion balance resolution).
• Scaled prototype goal: 1×10■■ N with increased emitter area and drive voltage.
• Thrust■to■power reporting normalized to a photon benchmark (F = P/c).
• Vacuum operation ≤ 1×10■■ Torr; hours■long drift■controlled integrations.

System Architecture

The system comprises four coherently coupled subsystems: 1) HF High■Voltage Emitter (Tesla■style resonant transformer, solid■state driven), providing intense, tunable EM fields. 2) Phase■Locked Multi■Coil Array (3■phase or N■phase), generating controlled rotating/standing field patterns. 3) Ground■Referenced Resonator (mini■Wardenclyffe), enabling Earth■coupled resonance experiments. 4) Metrology & Controls Stack: vacuum torsion balance, optical readout, photon calibrator, PLL/phase control, and clean cabling.

Figure 1. System Block Diagram



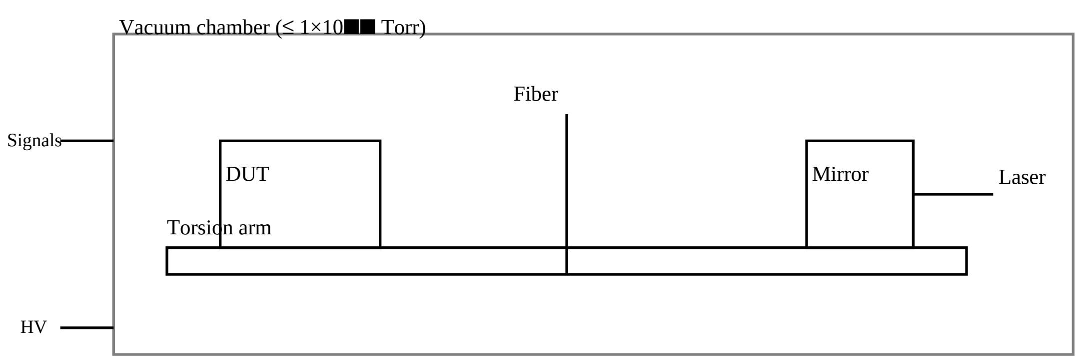
Bill of Materials (Core Items)

Subsystem	Component	Spec (typical)	Notes
Emitter	Secondary coil	Ø 100–150 mm, 800–1500 turns, AWG 28–34	Layered winding, varnish, toroidal topload
Emitter	Primary coil + cap	3–10 turns + 1–50 nF HV	Tune to secondary Fres; solid-state driver preferred
Emitter	HV supply	0–20 kV, ≥ 200 W	Current-limited; RF capable
Array	3 × coil formers	Ø 80–120 mm, 200–500 turns	Orthogonal layout for rotating field
Array	Phase driver	PLL/microcontroller (τ control)	120° (or programmable) phase offsets
Resonator	Ground plate	≥ 0.5 m² copper/aluminum	Low impedance connection to earth/chamber ground
Resonator	Topload sphere/toroid	100–300 mm	Smooth edges to suppress corona
Metrology	Vacuum chamber	≤ 1×10■■ Torr	Bake-out capability, RF feedthroughs
Metrology	Torsion balance	nN resolution	Optical lever or interferometric readout
Metrology	Photon calibrator	Laser + mirror (P/c)	On-arm reference force standard
Safety	μ■metal shields	As required	Magnetic cleanliness
Safety	Ceramic standoffs, interlocks	HV-rated	Arc protection, E■stop, bleed resistors

Wiring & Layout Notes

• Coaxial or twisted■pair returns for all high■current paths; mirror cable routes to enforce symmetry.
• Nonmagnetic fasteners; μ■metal shielding around transformers and drivers; optional Helmholtz coils for bias sweeps.
• Smooth, rounded HV electrodes to suppress corona; use RC snubbers and bleed resistors; interlocked enclosures.
• Dedicated star ground for measurement electronics; separate RF ground for emitter to minimize ground■loop forces.

Figure 2. Measurement Layout (Vacuum, Photon Calibrator, DUT on Torsion Arm)



Test Protocol (SOP)

1) Vacuum & Thermal: Pump to ≤ 1×10■■ Torr. Bake/soak until drift < 5 nN/hr. Record pressure vs time.
2) Null Baselines: With device off, log drift for ≥ 1 hr. Run dummy load (same power dissipation, no fields).
3) Photon Calibration: Apply laser to mirror; verify observed $F \approx P/c$ within ±5%. Use this to validate sensitivity.
4) Frequency Sweep: At fixed voltage, sweep emitter and array resonance. Record thrust, phase, and coherence metric.
5) Phase/Delay τ Sweeps: Vary phase offsets (0–360°) and delays; seek sign■reversing forces and lock■in patterns.
6) Orientation & B■field Sweeps: Rotate rig (0°, 90°, 180°) and vary Helmholtz bias. A real thrust tracks device phase; artefacts track orientation/field.
7) Thermal Controls: Replicate heating with electrical dummies. Confirm that any signal disappears without EM fields.
8) Replication: Repeat across days; then swap chambers/stands. Package raw logs and CAD for external replication.

Data Products & Acceptance Criteria

• Publish thrust■per■power (N/W) vs frequency and phase. Include vacuum, temperature, and magnetic logs.
• Acceptance threshold for anomaly: ≥ 5σ above combined noise, present in vacuum, sign■reversing with phase, absent in dummy runs.
• Report resolution limits (nN), integration windows, and uncertainty budget (thermal, magnetic, cable, outgassing, readout).
• Compare against photon benchmark and solar■sail equivalent to contextualize magnitude.

Physics Mapping (RCDEPS ↔ Hardware Controls)

• Drive term $\beta \cdot \sin(2\pi f t)$: Signal generator amplitude and frequency; array coil current set■point.
• Recursive term $\lambda \cdot E(t-\tau)$: Digital delay/feedback path into driver; sweep τ to map coherence memory effects.
• Phase field $\phi(r,t)$: PLL locks and phase offsets across coils; ϕ■dependent patterns realized by rotating/standing waves.
• Vacuum coupling observable: net force on torsion arm under phase/delay modulation in hard vacuum.

Safety & Compliance

High voltage and RF can be lethal. Use interlocked enclosures, current limiting, grounded shields, and E■stops. Observe RF exposure limits; add RF gaskets and filters on all feedthroughs. For vacuum glassware, add shields. Establish HV lockout/tagout procedures and PPE requirements. Keep detailed hazard analysis in the lab binder.

Appendix A — Initial Parameter Ranges

• Emitter resonance: 10–500 kHz (start at device's natural ω■; measure with low■power network analyzer).
• Array phase: 120° for 3■phase rotation; expand to programmable N■phase for pattern studies.
• Delay τ: 0–50 ms (low■Hz envelope coherence) and 0–500 µs (RF■scale recursion); step logarithmically.
• Voltage: 1–10 kV initial (air■free), 10–20 kV for scaling studies; increase area (A) per thrust scaling $F \propto A \cdot V^2 \cdot \eta$.

Appendix B — Go/No■Go Checklist

Item	Go Condition
Vacuum integrity	≤ 1×10■■ Torr for ≥ 2 hr; stable pressure

Item	Go Condition
Thermal drift	< 5 nN/hr baseline drift before powered tests
Photon calibrator	Measured F within ±5% of P/c benchmark
Cable symmetry	Coax/twisted pairs; mirrored return; no net loop areas
Magnetic cleanliness	μ■metal in place; Helmholtz bias mapped; ferrous parts minimized
Corona suppression	Rounded electrodes; no glow in dark-room check at test voltage
Dummy runs	Null within noise with resistive dummy; no ‘thrust’ under identical heating
Orientation tests	Signal independent of orientation; phase■flip reverses force sign

End of Manual — v1.0